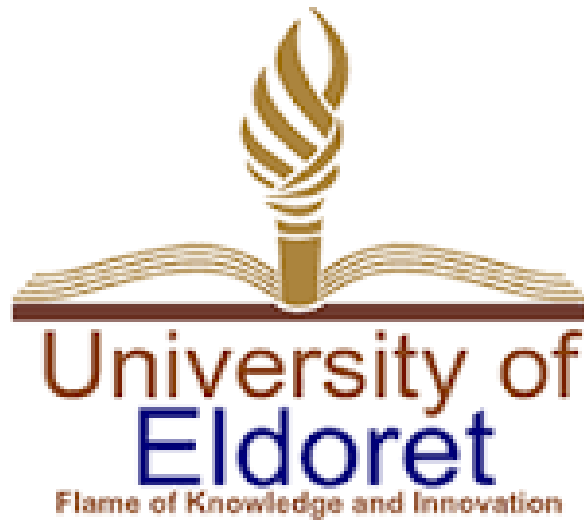




INSTITUTION: UNIVERSITY OF ELDORET

DEPARTMENT : OF MATHEMATICS AND COMPUTER SCIENCE

SCHOOL: OF SCIENCE

PROJECT TITLE

MACHO YA RAIA

BY

- 1) MIADING WUOR MIADING :COM/1003/24
- 2) ANTONY THAMAINI: COM/093/23
- 3) HIRAM SIMIYU: COM/1004/23
- 4) EDWARD LIANDA: COM/1007/24
- 5) LILIAN ATIENO: COM/016/23

LECTURER

NAME : DR FRANCIS MUSEMBI

SIGNATURE:

Declaration

We hereby declare that this project report, titled **macho ya raia** , is our own original work and has not been submitted, in whole or in part, for any other degree or professional qualification. All sources used or quoted have been duly acknowledged by means of complete references.

Date: March 31, 2026

DEDICATION

This project we dedicate it to each member of our development team above and the supervisor of this project who has been of great unwavering support throughout the development journey for this project .

Acknowledgement

We would like to express our deepest gratitude to our supervisors and instructors for their invaluable guidance throughout the development of this project. Their technical insights into geospatial data and software architecture were instrumental in shaping the final product.

We also wish to thank my peers and our lecturers who participated in initial surveys and feedback sessions. Your input helped ensure that **macho ya raia** was designed with the end-user in mind. Finally, we are grateful to our family and friends for their unwavering support and encouragement.

Abstract

Rapid urbanization often leads to a disconnect between citizens and local authorities regarding infrastructure maintenance and public services. **Macho ya raia** is a map-based civic engagement platform designed to bridge this gap. This project report details the development of a mobile and web-based application that allows residents to report non-emergency civic issues—such as potholes, broken streetlights, or illegal dumping—using real-time GPS tagging and photo documentation. By utilizing a full-stack architecture featuring a robust database and a responsive map interface, the system categorizes reports and visualizes them for both public awareness and administrative action. The goal of macho ya raia is to foster transparency, improve governmental accountability, and empower communities to take an active role in urban upkeep.

TABLE OF CONTENTS

PRELIMINARY PAGES

• Title Page	i
• Declaration	ii
• Dedication	iii
• Acknowledgement	iv
• Abstract	v
• Table of Contents	vi
• List of Tables	vii
• List of Figures	viii
• List of Abbreviations/Acronyms	ix

CHAPTER 1: INTRODUCTION

• 1.1 Project Overview	1
• 1.2 Background of the Study	1
• 1.3 Problem Statement	2
• 1.4 General Objective	2
• 1.5 Specific Objectives	3
• 1.6 Research Questions	3
• 1.7 Justification / Significance of the Study	
4	
• 1.8 Scope of the Project	4

CHAPTER 2: LITERATURE REVIEW

- 2.1 Introduction 6
- 2.2 Conceptual Review of Civic Technology
6
- 2.3 Review of Existing Civic Reporting Systems
7
- 2.4 The Role of GIS and Mapping in Civic Engagement 9
- 2.5 Review of Web and Mobile Technologies (PostgreSQL, Supabase, APIs)
10
- 2.6 Knowledge Gap Analysis 11

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

- 3.1 Introduction
13
- 3.2 Methodology (Software Development Life Cycle - SDLC)
13
- 3.3 Requirements Gathering and Analysis
14
 - 3.3.1 Functional Requirements
 - 3.3.2 Non-Functional Requirements
- 3.4 Feasibility Study (Technical, Economic, Operational)
15
- 3.5 System Modeling 16
 - 3.5.1 Use Case Diagram

• 3.5.2 Entity Relationship Diagram (ERD)	
• 3.5.3 Data Flow Diagrams (DFD Level 0 & 1)	
• 3.6 Architectural Design	18
• 3.7 Database Design and Schema	19

CHAPTER 4: IMPLEMENTATION AND TESTING

• 4.1 Introduction	21
• 4.2 Development Environment (Tools and Technologies)	21
• 4.3 Module Implementation	22
• 4.3.1 Frontend Development (User Interface)	
• 4.3.2 Backend Integration (Supabase/PostgreSQL)	
• 4.3.3 Map Integration (Leaflet/Google Maps API)	
• 4.4 System Testing	24
• 4.4.1 Unit Testing	
• 4.4.2 Integration Testing	
• 4.4.3 User Acceptance Testing (UAT)	

CHAPTER 5: DISCUSSIONS, CONCLUSION, AND RECOMMENDATIONS

• 5.1 Introduction	27
--------------------------	----

• 5.2 Discussion of Findings	27
• 5.3 Challenges Encountered	28
• 5.4 Conclusion	29
• 5.5 Recommendations for Future Work	30

REFERENCES

• (APA Style Citations)	32
-------------------------------	----

APPENDICES

• Appendix A: System User Manual	34
• Appendix B: Sample Source Code Snippets	36
• Appendix C: Project Schedule (Gantt Chart)	38
• Appendix D: Data Collection Instruments (Questionnaires/Interviews)	

Chapter 1: Introduction

1.1 Background Information

In the modern digital era, the concept of a "Smart City" relies heavily on the flow of information between a city's infrastructure and its inhabitants. However, many municipal feedback loops remain antiquated, relying on phone calls or physical visits to government offices. Civic technology, or "Civic Tech," seeks to leverage software to enhance the relationship between the public and the state. **Macho ya raia** emerges as a solution within this domain, focusing on crowdsourced data to identify and resolve local infrastructure failures.

1.2 Problem Statement

Currently, many citizens feel disempowered when encountering local issues like damaged public property or environmental hazards. The barriers to reporting these issues are high:

- **Lack of Centralization:** Uncertainty about which department handles specific problems.
- **Lack of Transparency:** Citizens rarely receive updates on whether their report was seen or acted upon.
- **Geospatial Inaccuracy:** Verbal descriptions of locations are often imprecise, leading to delays in maintenance. Without a streamlined, visual, and public-facing system, minor repairs escalate into costly infrastructure failures, and civic trust erodes.

1.3 Objectives

The primary objectives of macho ya raia project are:

- **To develop a user-friendly map interface** that allows citizens to drop pins on specific locations where issues occur.

- **To implement a categorization system** (e.g., Sanitation, Roads, Utilities) to ensure reports reach the correct authorities.
- **To provide a public dashboard** where users can track the status of reported issues (e.g., Pending, In Progress, Resolved).
- **To facilitate data-driven decision-making** for local governments by highlighting "hotspots" of recurring issues.

1.4 Justifications

Traditional reporting methods are inefficient and labor-intensive for both the public and the government. Macho ya raia justifies its development through:

- **Efficiency:** Automated tagging and categorization reduce administrative overhead.
- **Accountability:** Publicly visible reports encourage authorities to resolve issues timely.
- **Community Empowerment:** It transforms residents from passive observers into active stakeholders in their neighborhood's well-being.

Chapter 2: Literature Review

2.1 Evolution of Civic Technology

Civic tech has transitioned from simple informational websites to "Web 2.0" platforms that prioritize user-generated content. Early efforts focused on transparency (open data); modern efforts focus on "actionable" data—where user input directly triggers a service response.

2.2 Review of Existing Systems

Platforms like *FixMyStreet* in the UK and *SeeClickFix* in the US have set the standard for civic reporting. Research indicates that these platforms succeed because they reduce the "friction of participation." However, many international versions lack localization for specific regional infrastructures or fail to integrate with local database systems, creating a need for a bespoke solution like macho ya raia aka citizen eyes .

2.3 Geospatial Integration in Public Service

The use of Geographic Information Systems (GIS) is no longer restricted to urban planners. By integrating Google Maps or OpenStreetMap APIs, macho ya raia allows for "Spatial Crowdsourcing." Literature suggests that visually represented data is more easily interpreted by human managers than long lists of text-based complaints.

2.4 The Role of Crowdsourcing in Governance

Crowdsourcing serves as a "distributed sensor network." Instead of a city hiring thousands of inspectors to check streetlights, every citizen with a smartphone becomes a potential inspector. This model significantly lowers the cost of urban monitoring and increases the speed of incident detection.

Additional Issues & Considerations

- **Data Privacy:** The system must ensure that while reports are public, the personal identities of the reporters are protected according to local data protection laws.
- **Verification Mechanisms:** To prevent "spam" or false reporting, the project will explore the use of community voting (upvoting an issue to confirm it exists) and AI-based image verification.

- **Scalability:** The architecture must be capable of handling increased traffic during weather-related emergencies or large-scale infrastructure failures.